

DocentAI: Teacher-Guided AI for Personalised Education

1. Problem statement

The rapid evolution of technology, and particularly artificial intelligence (AI), is transforming the way knowledge is created, accessed, and applied. However, educational systems are not evolving at the same pace. This growing gap is generating a structural mismatch: while students live in highly dynamic, technology-driven environments, they are still often taught through methodologies that remain static, standardised, and increasingly outdated. As a result, many students struggle to engage with traditional learning models that do not reflect the way they process information today. At the same time, educators face increasing pressure to integrate new technologies into their teaching without having adequate tools that are aligned with pedagogical objectives. This situation leads to inefficiencies in learning, reduced motivation, and a progressive disconnection between education and real-world skills.

Although AI has emerged as a potential solution, most existing AI-based educational tools rely on general-purpose models trained on large-scale internet data. These systems are not aligned with specific curricula, lack pedagogical supervision, and often provide content that is inconsistent with classroom objectives. Furthermore, current solutions do not fully capture individual learning behaviours, such as recurring mistakes or differences in learning pace and style. There is therefore a clear need for a new generation of educational technology that bridges the gap between innovation and pedagogy. Such a solution must not only leverage the capabilities of AI but also ensure alignment with curricula, active teacher involvement, deep personalisation, and transparency in decision-making. This project is designed to respond directly to that need.

By addressing this gap, the proposed approach has the potential to generate significant educational, social, and economic impact. It can improve learning outcomes by enabling truly personalised educational experiences, increase student engagement by aligning learning methods with contemporary digital environments, and empower teachers with data-driven insights to support more effective decision-making. At a broader level, it contributes to reducing the disconnect between education and the demands of a rapidly evolving society, while also promoting more equitable access to high-quality learning tools. In doing so, the project anticipates the future direction of education in an increasingly technology-driven world.

2. Proposed solution

The proposed solution, DocentAI, is a teacher-guided AI system designed to deliver personalised learning experiences while ensuring full alignment with formal curricula. Unlike most existing tools, which simply answer questions about documents, it goes further: detecting mistakes, adapting explanations to each student, and continuously tracking individual progress. DocentAI is not designed to replace teachers, but to turn them into augmented educators – expanding what they can do. AI handles personalisation and feedback at scale, while the teacher retains control, judgment, and educational purpose.

The development of DocentAI will follow a structured and iterative approach, beginning with the design of a minimum viable product (MVP) focused on a specific subject area and educational context. A central component of the system is a controlled knowledge base, where teachers actively select, curate, and validate materials such as textbooks and course content. These resources are integrated through a retrieval-based architecture, ensuring that all AI-generated responses remain consistent with curriculum objectives. The platform will be built using scalable cloud infrastructure and a modular architecture, allowing future expansion across subjects, educational levels, and geographic contexts.

From a functional perspective, the system provides personalised and adaptive support to students while generating actionable insights for teachers. For instance, when a student repeatedly makes errors in isolating variables in algebra exercises, the system detects this pattern and delivers targeted explanations and tailored practice tasks. At the same time, the teacher dashboard aggregates these insights across students, enabling timely and data-informed classroom interventions. During exam preparation, students receive adaptive quizzes that dynamically adjust in difficulty and provide immediate, specific feedback, improving both efficiency and confidence.

The effectiveness of the solution will be evaluated through a set of measurable key performance indicators aligned with both learning and teaching objectives. Learning outcomes will be assessed through improvements in pre- and post-test scores, mastery rates, and reductions in recurring errors. Student engagement will be measured through interaction frequency, task completion rates, and time on task.

Teaching effectiveness will be evaluated based on reduced workload, faster feedback cycles, and the system's ability to identify at-risk students. Adoption metrics will include user satisfaction, perceived usefulness, and trust in AI-generated support.

In the current competitive landscape, solutions such as ChatGPT-based tools, Khan Academy's AI tutor, Duolingo's adaptive learning systems, and other EdTech platforms offer personalised learning experiences. However, most of these solutions rely on general-purpose models and lack direct teacher control over the content, which limits their alignment with formal curricula. The key differentiating factor of this project lies in its teacher-centred approach. By placing educators at the core of the system, both in content validation and pedagogical supervision, the solution ensures a level of reliability, transparency, and alignment that is not commonly found in existing tools – positioning it as a specialised and complementary solution for formal education.

3. Team and expertise

The successful development of this project relies on a multidisciplinary team that brings together complementary expertise across different fields. The team is composed of five core members whose combined knowledge enables the creation of a solution that is technically robust, educationally effective, user-centred, and economically sustainable. From a technical perspective, Miguel (software engineering) leads the design and development of the platform, including system architecture and AI integration, while Denisa (systems engineering) ensures infrastructure reliability, scalability, and efficient system performance. The pedagogical dimension is guided by Génesis (education), who designs evidence-based learning pathways and adapts the system to diverse educational contexts. She is supported by Laura (behavioural sciences), who provides insights into cognitive processes, motivation, and adaptive learning design. Finally, Mariano (economics) defines the economic and strategic framework, focusing on cost structures, business models, and long-term sustainability.

4. Budget

The development and first operational year of DocentAI is estimated at a total budget of \$3,483, of which \$3,133 covers planned expenditure and \$350 (~11% of planned costs) is held as a contingency reserve. The table below details all resources carrying an associated cost, organised across six categories. Free-tier tools used throughout the project — including GitHub, Cloudflare, Notion, Tally, Microsoft Clarity, and Framer — are excluded from this breakdown, as they do not contribute to the overall cost.

Item	Detail	Annual cost
1. Language models		\$729
GPT-5.4	5.4M tokens/mo (60 students x 15 msg/day x 200 tok/msg) \$2.5/M input + \$15/M output; \$60.75/mo (30/70 split)	\$729
2. Cloud infrastructure		\$660
Supabase Pro	PostgreSQL (file storage with auth and pgvector); \$25/mo	\$300
Vercel Pro	Frontend deployment, serverless functions; \$20/mo	\$240
Railway Hobby	Backend for API processing; \$5/mo	\$60
AWS S3	Curriculum materials storage; ~\$5/mo	\$60
3. Development tooling and monitoring		\$312
Sentry Developer	Error tracking, performance monitoring; \$26/mo	\$312
4. UX/UI design		\$432
Figma Professional	Design and prototyping; \$16/mo (1 active editor)	\$192
Assets and iconography	Icon libraries, illustration packs	\$60
Miscellaneous design	Onboarding flows, teacher analytics dashboard	\$180
5. Pedagogical validation and outreach		\$610
Participation incentives	Compensation for participants (2 validation cohorts)	\$240
Travel and logistics	Transport and costs for workshops and presentations	\$250
Materials and outreach	Printed documentation, branded materials	\$120
6. Miscellaneous operations		\$390
Domain	Custom registration, annual renewal	\$20
Resend	Email notifications; Free first 6 months, \$20/mo thereafter	\$120
Security review (GDPR)	Audit of data handling and privacy practices	\$250
Contingency reserve (~11%)		\$350
TOTAL		\$3,483